ADVANCED MATERIALS

Supporting Information

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Optical Fibers Embedded with As-Grown Carbon Nanotubes for Ultrahigh Nonlinear Optical Responses

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Supporting Information

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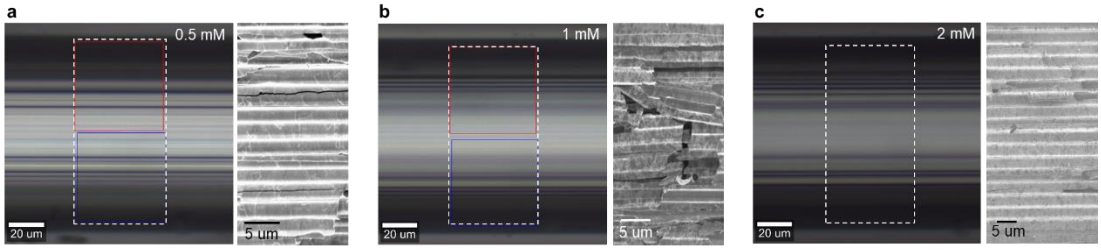


Figure S1. Different density of CNTs grown on the inner wall of PCF. a-c) Dependence of CNT density on the $FeCl_3$ catalyst precursor concentrations of 0.5 mmol L^{-1} (a), 1 mmol L^{-1} (b) and 2 mmol L^{-1} (c), respectively.

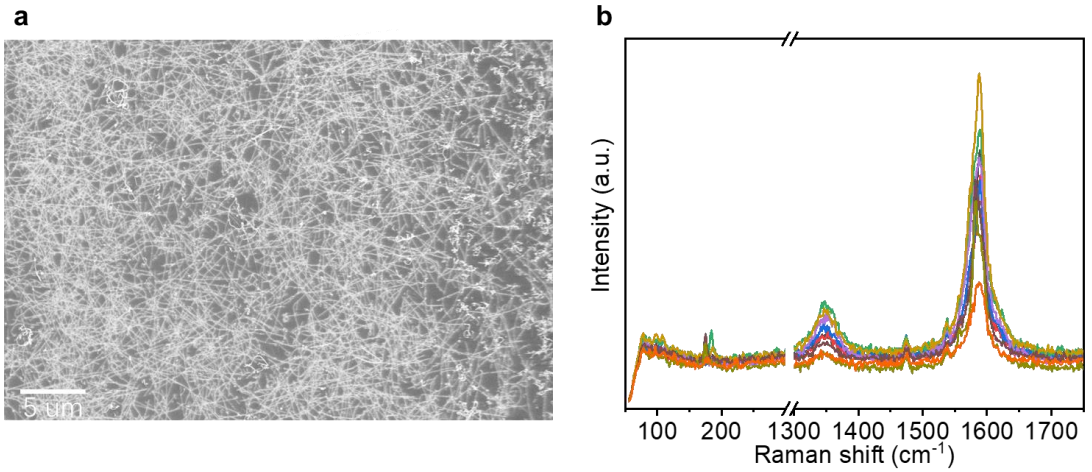


Figure S2. CNTs with poor quality grown on the inner wall of a PCF. a) SEM image of CNT embedded fibre grown by higher $FeCl_3$ catalyst precursor concentrations of 3 mmol L^{-1}. b) Raman spectra of the fibre embedded with CNT, using line mapping along radial direction with scan step of 3 μm, revealing the high D- to G-mode Raman intensity ratio.

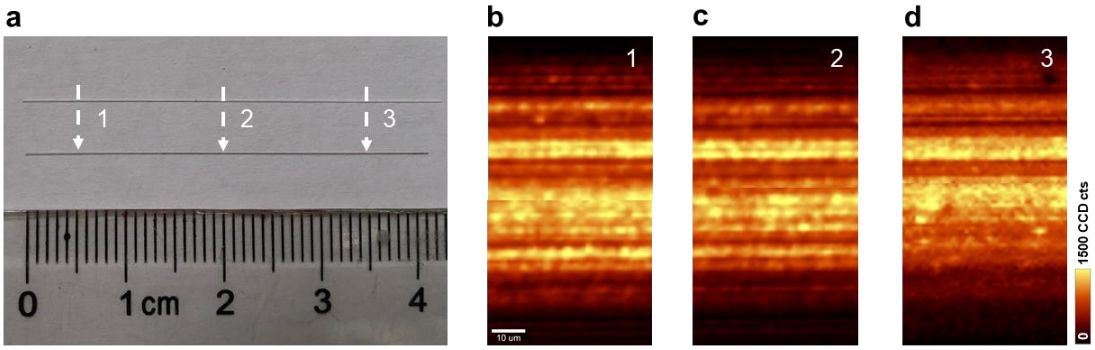


Figure S3. Uniform CNT growth into a 4-cm-long CNT embedded fibre. a) Typical photograph of the 4-cm-long fibre before and after CNT growth. b-d) G-mode Raman intensity mapping at the three positions of CNT embedded fibre labelled in (a), showing the uniformly distributed CNT film along the whole fibre.

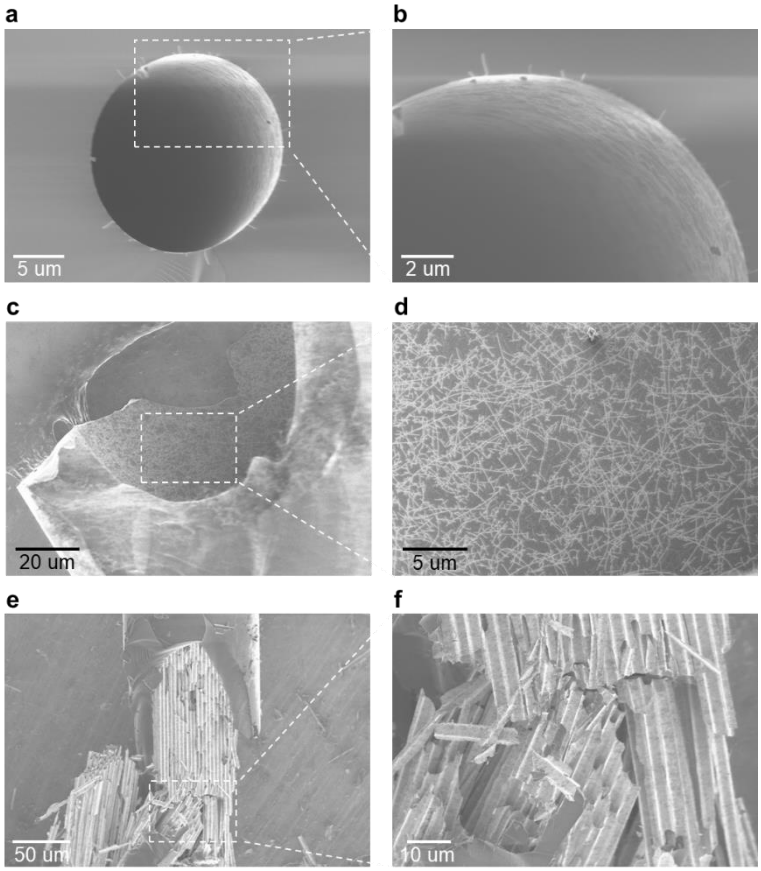


Figure S4. SEM images of CNT films at the fractured terminal of CNT fibres. a, c, e) Full coverage, randomly oriented CNT films grown on the inner wall of HCFs with hole diametres of 15 μm (a), 50 μm (c) and PCF (e). b, d, f) The zoom-in images of the CNT films in white dash square of (a, c, e), respectively.

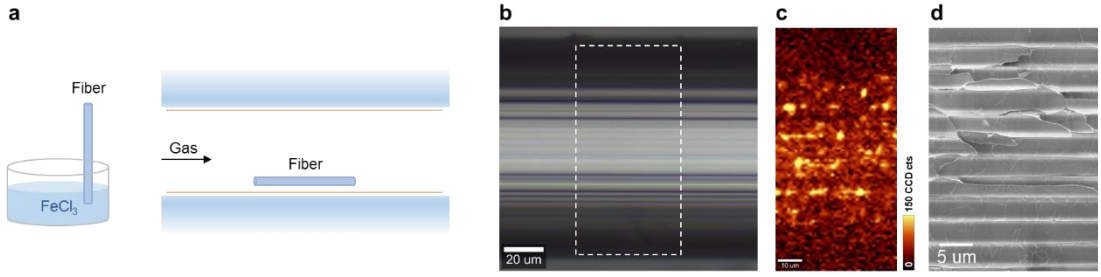


Figure S5. Controlled experiment of sending $FeCl_3$ into the fibres by capillary filling followed by high-temperature growth. a) Schematic of sending $FeCl_3$ into the fibres by capillary followed by high-temperature growth. b) optical image of as-grown CNT-PCF. c) G-mode Raman intensity mapping in white dash square of (b), showing sporadically distributed CNTs in the inner wall of PCF. d) SEM image of CNT films on the corresponding fractured CNT-PCF in (b), verifying the low density.

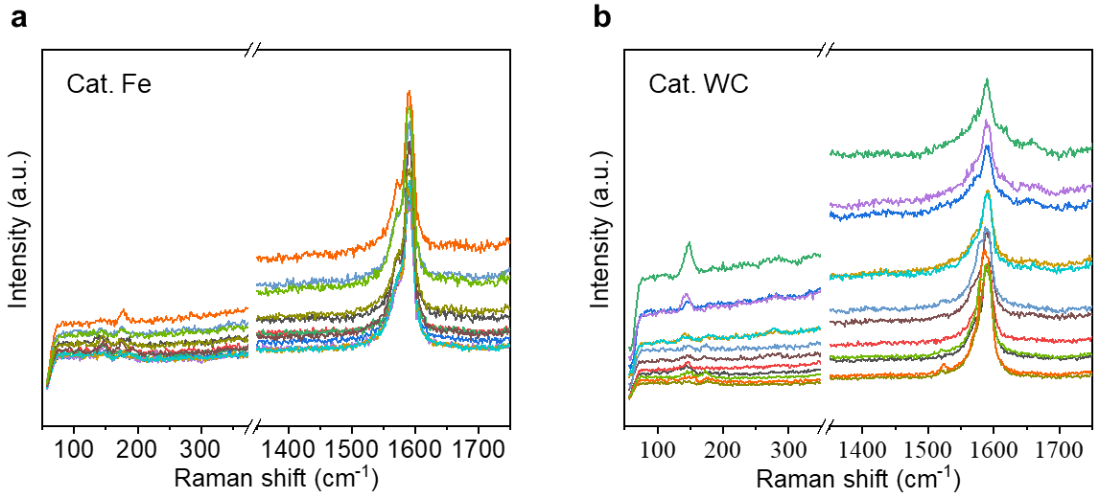


Figure S6. Raman spectra of the CNT-PCF using line mapping along radial direction with scan step of 3 μm and an excitation laser of 532 nm. a, b) Raman spectra of as-grown CNT-PCF by Fe (a) and WC (b) catalyst, indicating the CNT diametres of ~1.36 nm (a) and ~1.69 nm (b), respectively.

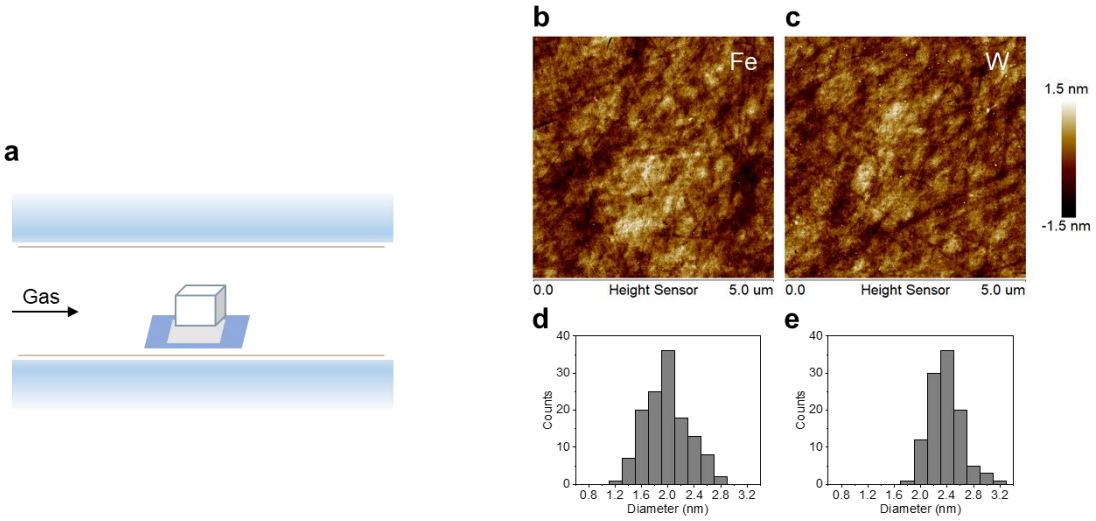


Figure S7. Diametre distribution of different catalyst by spatially confined CVD method monitoring the CNT growth conditions in an optical fibre. a) Schematic of spatially confined CVD method using quartz-Si/SiO_2 stacking with a cubic weight. b, c) AFM image of quartz loaded with catalyst precursor Fe (b) and W (c). d, e) Diametre distribution of catalyst nanoparticles Fe (d) and W (e), and the average diametre of Fe nanoparticles (~2.0 nm) is smaller than that of W nanoparticles (~2.4 nm).

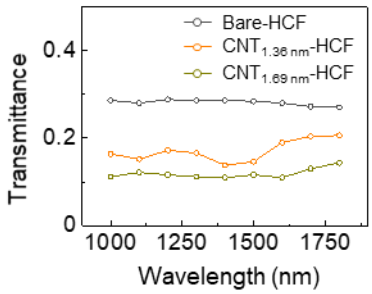


Figure S8. The transmittance of bare-HCF and CNT-HCF with ~1.36 nm and ~1.69 nm-diametre CNTs. The CNT-HCF with tube diametre of ~1.36 nm exhibited low transmittance near 1500 nm, which corresponds to its S_{11} absorption peak. For CNT-HCF with tube diametre of ~1.69 nm, no obvious peak was observed in the range of 1000~1800 nm, because the position of S_{11} absorption peak was beyond the measured range, and the characteristics of S_{22} and M_{11} were weak. There is a large coupling loss in HCFs, because the numerical aperture of the objective lens is not perfect match with the core size.

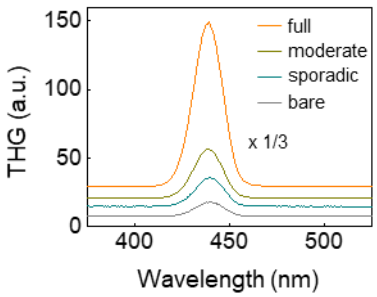


Figure S9. THG enhancement performance of CNT-HCF with different CNT coverage. We provide the THG signal intensity of HCF embedded with different coverage of CNTs described in Figure 1. The peak intensity ratio of the four samples (bare, sporadic, moderate and full coverage CNTs-HCF) is about 1:2:4:33.

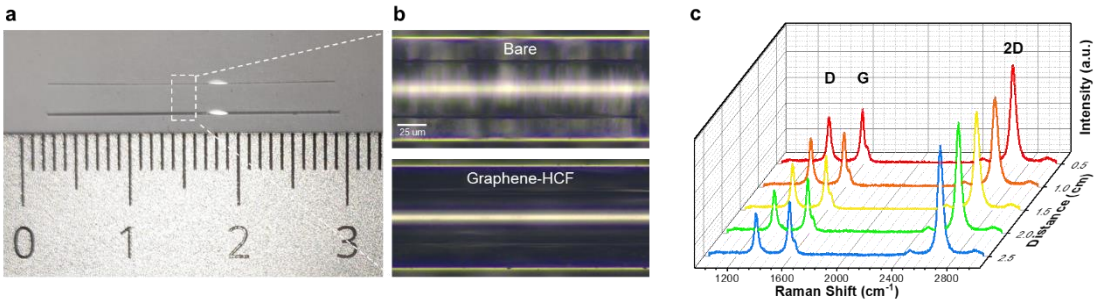


Figure S10. Uniform graphene growth into a 3-cm-long optical fibre. a) Typical photograph of the 3-cm-long fibre before and after graphene growth. b) Magnified optical image in white dash square of (a), showing the graphene grown into an HCF with a core diametre of ~50 μm. c) Raman spectra of the graphene-HCF in (a). The D-, G- and 2D-mode Raman peaks of the graphene at different positions exhibit little variation, indicating the uniform graphene growth along the whole fibre.

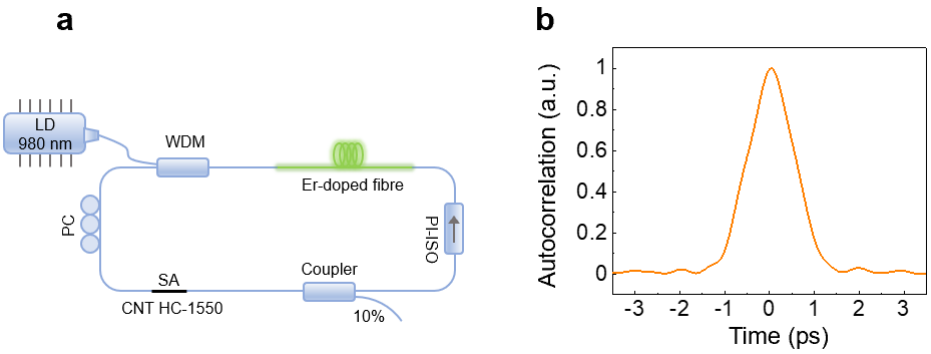


Figure S11. Stretched soliton generated by CNT-embedded HC-1550 based mode-locked ring fibre laser. a) Schematic diagram of an ultrafast fibre laser operating at 1561 nm. b) Autocorrelation trace of 1561 nm fibre laser output without external cavity dispersion compensation with 1.23 ps FWHM.

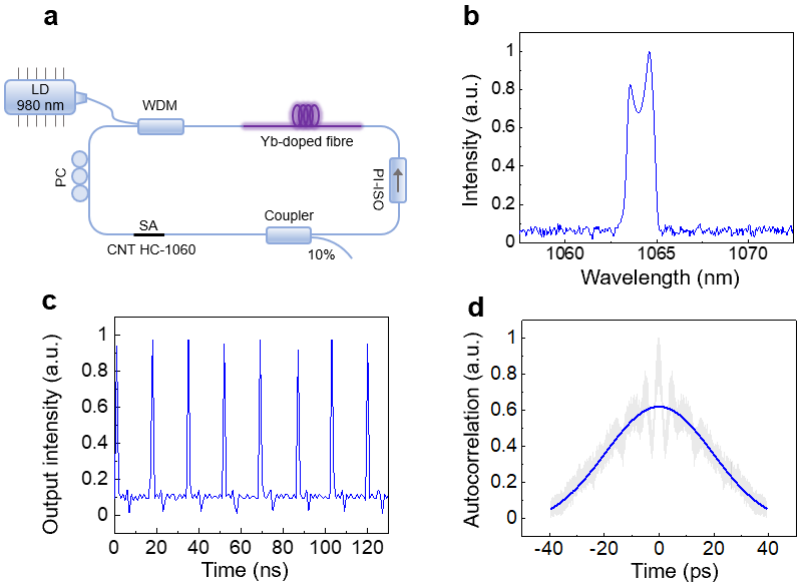


Figure S12. Dissipative soliton generated by CNT-embedded HC-1060 based mode-locked ring fibre laser. a) Schematic diagram of an ultrafast fibre laser operating at 1064 nm. b) Output spectrum with a central wavelength at 1064 nm. c) Pulse train with ~17 ns interval (~58.8 MHz). d) Autocorrelation trace with Gaussian fit FWHM of 46.1 ps.